$$\int \frac{1}{\sqrt{x^2+x}}\,dx$$

$$x^2+x = \underbrace{\left(x+\tfrac{1}{2}\right)^2}_{x^2+x+\frac{1}{4}} - \frac{1}{4}$$

$$\int \frac{1}{\sqrt{\left(x+\frac{1}{2}\right)^2-\frac{1}{4}}}\,dx$$

$\sqrt{x^2-a^2}$ $\quad x = a\sec\theta$

or sub $u = x+\frac{1}{2}$

$$x+\frac{1}{2} = \frac{1}{2}\sec\theta \qquad dx = \frac{1}{2}\sec\theta\tan\theta\,d\theta$$

$$\sqrt{\left(x+\frac{1}{2}\right)^2-\frac{1}{4}} = \sqrt{\frac{1}{4}\sec^2\theta-\frac{1}{4}} = \frac{1}{2}\tan\theta$$

$$\int \frac{1}{\frac{1}{2}\tan\theta}\,\frac{1}{2}\sec\theta\tan\theta\,d\theta$$

$\int \sec\theta \, d\theta$

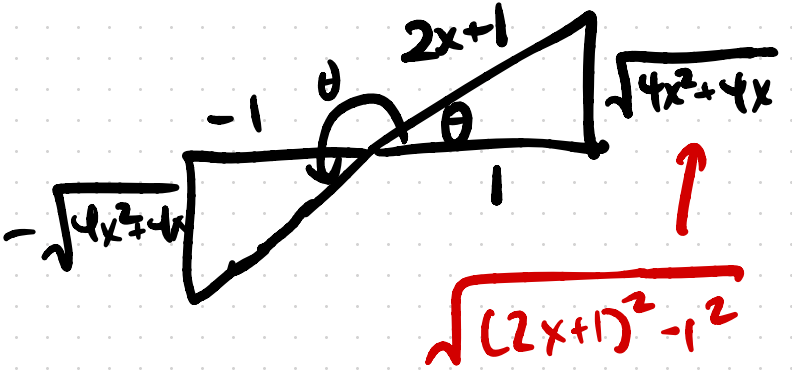
$0 \le \theta < \frac{\pi}{2}$ or $\pi \le \theta < \frac{3\pi}{2}$

$\ln|\sec\theta + \tan\theta| + C$

$x + \frac{1}{2} = \frac{1}{2}\sec\theta$

$2x+1 = \sec\theta = \frac{\text{hyp}}{\text{adj}}$

$\ln\left|2x+1+\sqrt{4x^2+4x}\right| + C$



Integration of Rational Functions by Partial Fractions (7.4)

A **rational function** is a ratio of polynomials:

$$\frac{P(x)}{Q(x)}, \quad \text{where } P \text{ and } Q \text{ are polynomials.}$$

The method of **partial fractions** gives a way to evaluate the integral of any rational function.

Fact: Any rational function $P(x)/Q(x)$ can be written as

$$\frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}, \quad \deg R < \deg Q$$

using polynomial division.

degree of polynomial

$$\int \frac{x^2+x+5}{x+2}\,dx$$

deg numerator $\geq$ deg denominator
So we should first divide.

$$\begin{array}{r} x-1 \\ x+2 \overline{)\, x^2+x+5} \\ \underline{x^2+2x} \\ -x+5 \\ \underline{-x-2} \\ 7 \end{array}$$

$$\frac{x^2+x+5}{x+2} = x-1+\frac{7}{x+2}$$

$$\int \frac{x^2+x+5}{x+2}\,dx = \int \left(x-1+\frac{7}{x+2}\right)dx$$

$$= \frac{1}{2}x^2 - x + 7\ln|x+2| + C$$

Fact: Any real polynomial can be factored into linear and irreducible quadratic factors:

$$P(x) = c(x-r_1)(x-r_2)\cdots(x^2+a_1x+b_1)(x^2+a_2x+b_2)\cdots$$

c: leading coefficient

$(x^2+a_1x+b_1)$: irreducible, i.e. no real roots, i.e. $a_1^2 - 4b_1 < 0$

Some factors may repeat.

Fact: If P, Q are real polynomials with $\deg P < \deg Q$, and

$$Q(x) = (x-r_1)^{e_1}(x-r_2)^{e_2}\cdots(x^2+a_1x+b_1)^{f_1}(x^2+a_2x+b_2)^{f_2}\cdots$$

then there are constants $A, B, C, \ldots$ such that

$$\frac{P(x)}{Q(x)} = \frac{A}{x-r_1} + \frac{B}{(x-r_1)^2} + \cdots + \frac{C}{(x-r_1)^{e_1}}$$

$$+ \frac{D}{x-r_2} + \frac{E}{(x-r_2)^2} + \cdots + \frac{F}{(x-r_2)^{e_2}}$$

$$\vdots$$

$$+ \frac{Gx+H}{x^2+a_1x+b_1} + \frac{Ix+J}{(x^2+a_1x+b_1)^2} + \cdots + \frac{Kx+L}{(x^2+a_1x+b_1)^{f_1}} + \cdots$$

Example:

$$\frac{x}{(x^4-1)^2}$$

Factor denominator into linear and quadratic factors.

$$x^4-1 = (x^2-1)(x^2+1) = (x-1)(x+1)(x^2+1)$$

$$\frac{x}{(x-1)^2(x+1)^2(x^2+1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1} + \frac{D}{(x+1)^2}$$

$$+ \frac{Ex+F}{x^2+1} + \frac{Gx+H}{(x^2+1)^2}$$

$$\frac{x}{(x^4-1)^2} = \frac{-1/8}{x-1} + \frac{1/16}{(x-1)^2} + \frac{-1/8}{x+1} + \frac{-1/16}{(x+1)^2}$$

$$+ \frac{(1/4)x}{x^2+1} + \frac{(1/4)x}{(x^2+1)^2}$$

Found using computer

$$\int \frac{x-12}{x^2-4x}\,dx$$

deg num < deg dem

So we don't need to do long division.

$x(x-4)$

$$\frac{x-12}{x(x-4)} = \frac{A}{x} + \frac{B}{x-4}$$

= : equality of functions

Find A and B:

Multiply both sides by $x(x-4)$

$$x-12 = A(x-4) + Bx$$

This must hold for all x.

$x=4$ $\quad -8 = A\cdot 0 + B\cdot 4$ $\quad B=-2$

$x=0$ $\quad -12 = A(-4) + B\cdot 0$ $\quad A=3$

optional double check

$$\frac{x-12}{x(x-4)} = \frac{3}{x} - \frac{2}{x-4} = \frac{3(x-4)-2x}{x(x-4)} = \frac{x-12}{x(x-4)}$$

$$\int \frac{x-12}{x(x-4)}\,dx = \int \left(\frac{3}{x} - \frac{2}{x-4}\right) dx$$

$$= 3\ln|x| - 2\ln|x-4| + C$$

$$\int \frac{x^2+x+1}{(x+1)^2(x+2)}\,dx$$ deg num < deg dem

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = \frac{A}{x+1} + \frac{B}{(x+1)^2} + \frac{C}{x+2}$$ Multiply by $(x+1)^2(x+2)$

$$x^2+x+1 = A(x+1)(x+2) + B(x+2) + C(x+1)^2$$

$x=-1$ $\quad (-1)^2+(-1)+1 = B\cdot 1$

$1 = B$

$x=-2$ $\quad (-2)^2+(-2)+1 = C\cdot(-1)^2$ $\quad C=3$

$x = 0$ $\quad 1 = A \cdot 1 \cdot 2 + B \cdot 2 + C \cdot 1^2$

$1 = 2A + 2 + 3$

$-2 = A$

OR

$x^2 + x + 1 = A(x^2 + 3x + 2) + (x + 2) + 3(x^2 + 2x + 1)$

$-2x^2 - 6x - 4 = A(x^2 + 3x + 2)$ polynomials on both sides must be the same

$-2 = A$

$$\frac{x^2+x+1}{(x+1)^2(x+2)} = -\frac{2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}$$

$$\int \frac{x^2+x+1}{(x+1)^2(x+2)}\,dx = \int \left(-\frac{2}{x+1} + \frac{1}{(x+1)^2} + \frac{3}{x+2}\right)dx$$

$$= -2\ln|x+1| - \frac{1}{x+1} + 3\ln|x+2| + C$$